

SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 3.1: Statistics I — Teacher Reference (Pre-filled)

SUMMARY TABLE: CORE MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 3.1: Statistics I
Driving Question	DRIVING QUESTION: How can we use statistics to tell an honest and useful story about data from our own Kenyan community — and help people make better decisions?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 What Is This Data Trying to Tell Us? — Launching the Maize Price Mystery	A chaotic list of 40 daily maize flour prices from a Wakulima Market stall, collected over 8 weeks. A family says prices vary wildly; their neighbour says prices are always about the same. Both shop at the same stall.	Raw, unorganised data cannot settle a real-life disagreement fairly. You can find the minimum and maximum price by scanning the list, but you cannot easily see patterns, typical values, or the overall shape of the data. To use data honestly and usefully, you need tools to organise and represent it.	Statistics is the science of collecting, organising, representing, and interpreting data to support honest decision-making. The Wakulima Market mystery cannot be solved by looking at raw numbers alone — we need frequency tables, measures of central tendency (mean, median, mode), and graphs (histograms, frequency polygons). These are the tools we will build in the coming lessons to finally settle the argument between the family and their neighbour.
Lesson 2 Meeting the Maize Price Data: What Is Raw Data and Where Does It Come From?	Learners examined a raw list of 40 maize flour prices from Wakulima Market and found it nearly impossible to determine — by eye alone — whether prices were stable or variable. They also observed a live raw data set of walk-to-school times collected from their own class, which presented the same challenge of unorganised numbers.	Raw data is data that has not yet been organised or summarised. Data can come from primary sources (collected directly, e.g., asking classmates) or secondary sources (e.g., a market price list). Tally marks are a tool for organising raw data into frequencies. A frequency is the count of how many times a value or group of values appears in a data set.	The reason both the Nairobi family and their neighbour could be telling the truth is that raw data — a jumbled list of numbers — does not reveal patterns clearly. Without organising the 40 maize prices, different people notice different numbers and reach different conclusions. To settle the argument fairly, we need tools that organise the data so the full picture becomes visible.
Lesson 3 Organising the Chaos: Frequency Distribution	Learners worked with the raw list of 40 daily maize flour prices from Wakulima Market — the same chaotic data from the anchoring	A frequency distribution table for ungrouped data organises raw data by listing each distinct value alongside a tally and	When we organised the 40 Wakulima Market maize prices into a frequency distribution table, we could identify the

Tables for Ungrouped Data	phenomenon — and organised it by tallying each price value and recording its frequency in a three-column table (Price Tally Frequency). They verified their table summed to 40 entries and compared their pair table to a class-constructed table on the board.	frequency count. The value with the highest frequency is the mode. Organising raw data into a frequency table reveals patterns — such as which prices are common and which are rare — that are completely invisible in the raw list. The sum of the frequency column must always equal the total number of data points (40 in our dataset).	modal price — the price that appeared most often. This gave us our first structured evidence to begin answering the driving question: the family who felt prices 'varied wildly' may have visited on low-frequency days at the extremes, while the neighbour who said prices were 'always about the same' may have experienced the high-frequency modal price cluster. The table alone cannot fully settle the argument — we still need measures of central tendency and a graphical display to tell the complete, honest story.
Lesson 4 Frequency Distribution Tables for Ungrouped Data — Bringing Order to the Maize Price Mystery	Learners observed the same 40 raw Wakulima Market maize prices from Lesson 1 being transformed — through a systematic tally process — into a frequency distribution table with four price values (KSh 98, 105, 112, 120), their tally marks, frequencies (16, 12, 8, 4), and a calculated Σfx column summing to $\Sigma fx = 4204$.	Learners learned that: (i) frequency means the number of times a value occurs in a data set; (ii) a frequency distribution table organises raw data into a structure that immediately reveals which values are common and which are rare; (iii) the mode is the value with the highest frequency; (iv) the mean of ungrouped data can be calculated as $\Sigma fx \div \Sigma f$, and this weighted mean (KSh 105.10) differs from a simple midrange estimate because it accounts for how often each price actually occurred.	Learners explained how the frequency table provides the first structured evidence in the Maize Price Mystery: the mode (KSh 98, occurring 40% of the time) supports the neighbour's view that prices are stable, while the presence of KSh 120 prices (occurring 4 times) supports the family's experience of occasional expensive days — showing that both parties were telling the truth from different statistical perspectives. Learners also explained why organising data with integrity (recording every value accurately) is essential for fair community decision-making.
Lesson 5 Histograms: Making the Maize Price Distribution Visible	Learners observed a teacher-modelled histogram of the 40 Wakulima maize flour prices (equal class-width), then observed a second demonstration showing how a histogram with unequal class widths MISLEADS when bar height = frequency rather than frequency density. Learners saw that the shape of the resulting histogram is right-skewed — with the modal class in the KSh 100–109 range and a thin tail extending toward KSh 140–149.	Learners learned that: (i) a histogram uses bars whose AREA represents frequency, so frequency density (frequency $\div$ class width) must be used on the y-axis when class widths are unequal; (ii) histograms have no gaps because the data (price) is continuous; (iii) the shape of a histogram (skew, spread, modal class) carries meaning — a right-skewed distribution explains why the neighbour experiences 'stable' prices on most days while the family occasionally encounters shockingly high prices on rare days; (iv) the same data can look different depending on how class intervals are chosen, raising questions about honest data representation.	Learners explained, in group Verdict Statements and individual Model Update Statements, that BOTH the Wakulima family and their neighbour are telling the truth: the histogram shows that most of the 40 days had prices clustered in the lower classes (supporting the neighbour's experience of stability), but the right-skewed tail confirms that occasional high-price days do exist and feel dramatic when experienced (supporting the family's experience of variability). The histogram provides stronger evidence than the mean, mode, or median alone because it shows the FULL SHAPE of the distribution, not just a single summary number.
Lesson 6 Grouping the Maize	Learners sorted 40 Wakulima Market maize price cards into class-interval bins using	Learners learned that: (a) grouping raw data into class intervals makes the	Learners explained, in written Data Story sentences and updated cumulative models,

Price Data: Grouped Frequency Tables and the Modal Class	three different class widths (Ksh 5, Ksh 10, Ksh 20) and constructed grouped frequency distribution tables by hand. They performed a gallery rotation comparing all three tables and independently grouped 24 months of Kisumu rainfall data into a second grouped frequency table.	distribution pattern visible but loses individual values; (b) the modal class is the class interval with the highest frequency — it is a range, not a single number; (c) the choice of class width changes the appearance of the data and can make the same dataset look either 'stable' or 'variable', directly affecting which side of the family-neighbour dispute is supported.	how the modal class of Ksh 50–59 for the maize price data relates to the ungrouped mode found in Lesson 3, and what this reveals about the typical price a Nairobi family pays at Wakulima Market. They connected the Kisumu rainfall modal class to practical farming decisions, and updated their Current Best Answer to the driving question to incorporate grouped data evidence.
Lesson 7 Measures of Central Tendency for Grouped Data — Cracking the Maize Price Mystery at Scale	Learners observed the grouped frequency distribution table of 40 Wakulima Market maize flour prices organised into 8 class intervals, completed the cumulative frequency column, and filled in a deviation table using an assumed mean of 72 Ksh.	Learners learned how to estimate the mean of grouped data using the assumed-mean/deviation method, identify the modal class and its midpoint as the estimated mode, and apply the median interpolation formula ($\text{Median} = L + \frac{(n/2 - F)/f}{h}$) to find the median class and estimated median value.	Learners explained that all three measures of central tendency — mean (≈ 72.7 Ksh), modal class midpoint (72 Ksh) and median (≈ 73.2 Ksh) — converge around 72–73 Ksh, consistent with the neighbour's experience of typical days, while the family's experience reflects rare high-price days and spread. The family's perceived variation must therefore be about the spread of prices around that centre, not about the centre itself — a question that drives the next lesson.
Lesson 8 Which Number Tells the Truth? — Final Explanation: Mean, Mode and Median of the Maize Price Data	Learners observed that when a single extreme price (KSh 45 mandazi spike; analogous to the Rift Valley-shortage maize price spike) is included in a data set, the mean is pulled noticeably higher than both the median and the mode — even though the extreme value occurred only once. They also observed, by completing the DQB Closure Ceremony, that every question posted across eight lessons now has an evidence-based answer.	The mean of ungrouped data ($\Sigma fx \div \Sigma f$) is sensitive to outliers and can misrepresent the 'typical' value experienced by most people. The median and mode are more resistant to extreme values. The choice of which measure to report is a deliberate decision with real consequences for how data tells its story — and for whether that story is honest. For the Wakulima Market maize price data: mean $\approx$ KSh 105.10, median $\approx$ KSh 103, and the mode reflects the most commonly charged price; the family budgeting for weekly shopping should use the median or mode for a realistic typical-day estimate.	Both the Nairobi family and their neighbour were telling the truth as they experienced it. The neighbour, shopping on typical days, encountered prices clustered near the mode and median — so prices felt 'always about the same.' The family, whose data included days with a Rift Valley supply-shortage spike, experienced genuine price variation — and their mean was inflated by those rare high-price days. Statistics resolves the mystery not by declaring one person wrong, but by showing that different measures capture different aspects of reality — and that choosing the right measure depends on the question you are trying to answer and the decisions that rest on it.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

Licensing & Copyright: © 2026 SeaVuria and ARES. This work is licensed under a [Creative Commons Attribution-NonCommercial 4.0 International License \(CC BY-NC 4.0\)](https://creativecommons.org/licenses/by-nc/4.0/). To view a copy of this license, visit <https://creativecommons.org/licenses/by-nc/4.0/>. You are free to copy, redistribute, and adapt this material in any medium or format, provided appropriate credit is given and the material is not used for commercial purposes. KICD learning outcomes and curriculum designations reproduced here for alignment purposes remain the property of KICD and are not covered by this license. Linked external resources (e.g. CK-12, Khan Academy, PhET) are subject to their own licenses.
